

JOEY SHI

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EXPERIENCE

Amazon

May 2025 – Present

Software Development Engineer

Vancouver, BC

- On a team building internal developer tools and AI agents used by engineers across Amazon.
- Re-architected an internal Slack knowledge gateway from sync to async with a dispatcher Lambda and SQS, cutting acknowledgement latency to 0.5s and eliminating timeout errors.
- Scaled an internal diagnostic CLI into a team-wide operations platform with multi-service/multi-region config, SSH log fetching, CloudWatch alarm integration, and stuck-thread detection.
- Integrated enterprise SSO into an internal API across beta, gamma, and prod using AWS CDK (Route 53, API Gateway custom domains), and authored the SOP teammates use to add auth to their endpoints.

Visier Inc.

April 2023 – May 2025

Software Developer II

Vancouver, BC

- On the data ingestion team for an HR analytics SaaS platform serving enterprise customers.
- Implemented and owned Scala data connectors for MySQL, BambooHR, Oracle Fusion, Medallia, Greenhouse, Redshift, and Snowflake; rapid delivery of the MySQL connector unblocked a new customer contract.
- Added limit/offset pagination to the MySQL connector, fixing gateway timeouts on customers with very large source tables.
- Built data flow graph visualization libraries in TypeScript and Angular to simplify ETL pipeline configuration.
- Owned on-call rotation for all connectors I shipped, diagnosing and resolving customer-impacting ingestion failures in production.

Vital Mechanics Research

May 2022 – Oct 2022

Software Developer

Vancouver, BC

- At a research lab building 3D clothing simulation software for the apparel industry.
- Developed a Python CLI tool for extracting garment geometry and stitch information from glTF and DXF files for 3D clothing simulation.
- Implemented mathematical optimization algorithms to model garment edges as Bézier curves and approximate stitch point positions along the garment.

PROJECTS

Differential Equation Solver | Python · Matplotlib · Tkinter

- Implemented finite difference algorithms to solve ordinary and partial differential equations, with a Tkinter GUI and animated Matplotlib plots for visualizing solutions.
- Built a CI/CD pipeline with PyTest, Tox, and GitHub Actions, and published the application on the Python Package Index.

PC Builder Website | Java · Angular · MySQL · Docker

- Built a full-stack web app for browsing PC builds and picking components, with a Java REST API, MySQL schema, and Angular frontend.
- Wrote docker-compose deployment scripts and self-hosted the app on a Linux server behind a Caddy reverse proxy.

Multiplayer Browser Platformer | TypeScript · Node.js · Socket.IO

- Built a real-time multiplayer browser platformer using the Canvas 2D API, with a WebSocket backend that broadcasts player state across clients.
- Implemented player movement, sprite animation, game physics, map rendering, and collision logic from scratch.

EDUCATION

University of British Columbia

Sep 2018 – May 2023

Bachelor of Science | Combined Major in Computer Science and Mathematics, with Distinction

Vancouver, BC

- Graduated with a 4.0/4.0 GPA.
- Received the Stanley M. Grant Scholarship in Mathematics (2021) and Trek Excellence Scholarship (2020).

TECHNICAL SKILLS

Languages: Python · Rust · Scala · Java · TypeScript · JavaScript · HTML5 · CSS3 · C/C++ · SQL · Zsh

Frameworks/Libraries: Angular · React · Node.js · D3.js · Flask · Pandas · Pytest · Javalin · ScalaTest · AWS CDK

Tools/Technologies: Lambda · SQS · DynamoDB · CloudWatch · Cloudflare · Docker · Git · Arch Linux · Neovim · tmux